

Chapter 1

Bernoulli's equation

Question: Describe the equation of conservation of mass on the equation of continuity.

Answer: Let us consider a stream filament of liquid in the steady motion of an inviscid fluid.

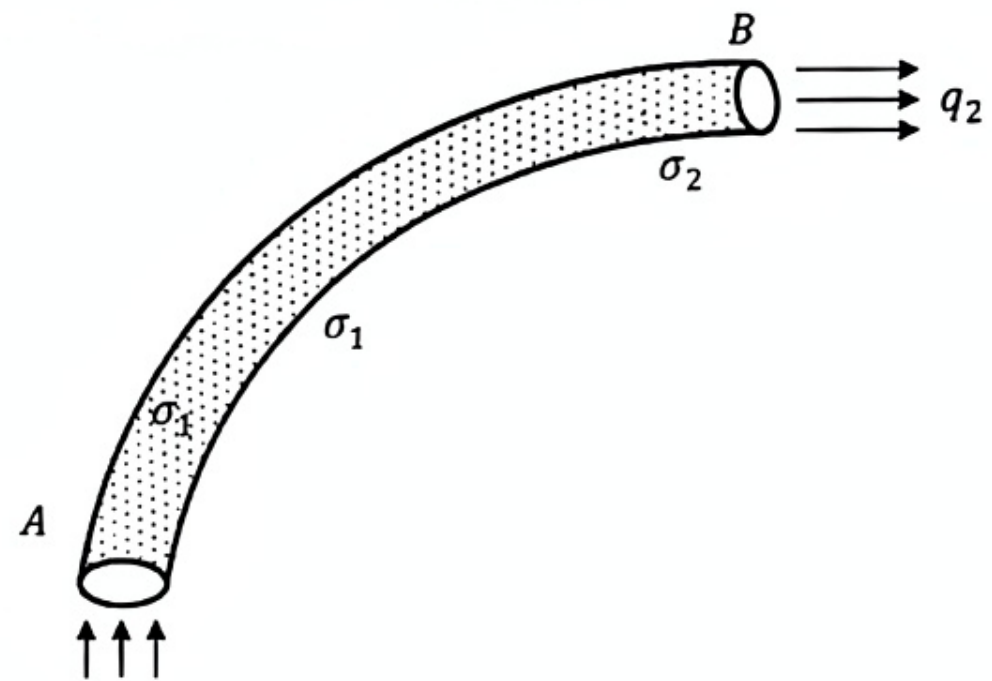
Let σ_1 and σ_2 be the cross-section areas of infinity small dimension of the fluid A & B respectively and q_1 & q_2 be the speeds of A & B respectively,

$$\text{Here, } q_1 = \frac{dx}{dt}$$

$$\text{or, } dx = q_1 dt$$

$$\text{or, } \int dx = \int q_1 dt$$

$$\text{or, } x = q_1 t \text{ [volume} = q_1 t \sigma_1 \text{]}$$



Then the volume of the fluid entering at A in the time is $q_1 t \sigma_1$ [$\therefore$ Volume=area $\times$ height]

So that the volume of the fluid in unit line is $q_1 \sigma_1$ at A. Similarly, $q_2 \sigma_2$ will be the volume in unit line going out at B. If ρ_1, ρ_2 be the density of fluid at A and B respectively. Then the mass of the fluid entering at A and going out through B will be $q_1 \rho_1 \sigma_1$ and $q_2 \rho_2 \sigma_2$. Since the motion is steady same mass of fluid must enter at A and go out at B.

We must have;

Now, if the fluid is incompressible $\rho_1 = \rho_2$

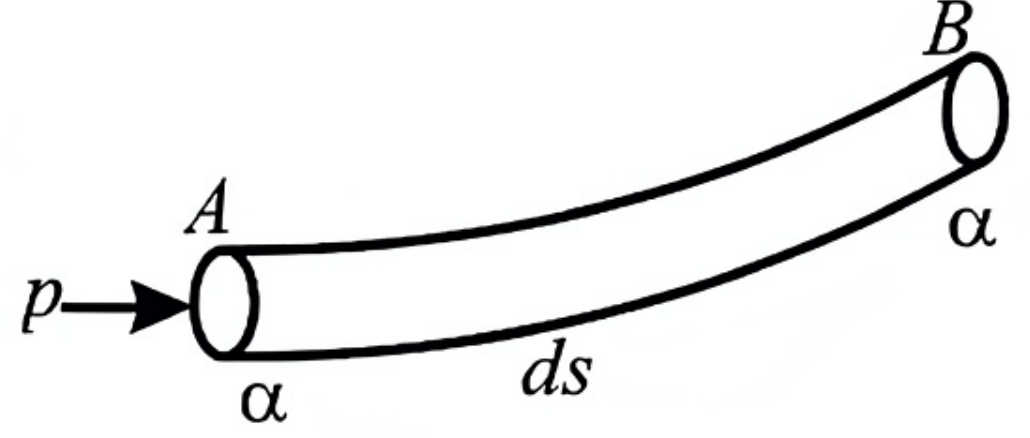
So that $q_1 \sigma_1 = q_2 \sigma_2$; which is called the equation of continuity or the equation of conservation of mass or simply $q\sigma = \text{constant}$

Question: State and prove Bernoulli's Theorem for inviscid fluid.

Statement: For the steady motion of an inviscid barotropic fluid under conservative body force, the pressure at a point is given by,

$$\int \frac{dp}{\rho} + \frac{1}{2}q^2 + V = \text{Constant} \quad (1)$$

Proof: Consider a short and circular cylindrical element AB of fluid whose axis is parallel to the flow at the element. Let the normal cross-sectional area of the cylinder be α and its length be ds where s is arc of the streamline on which the element lies, measured from a fixed point on it. The pressure p in the curved surface cancel out and hence contribute nothing to the resultant force in the direction of motion. Thrust on the end A is $p\alpha$ in the direction of motion, while on the other



end B, the thrust is $-\left(p + \frac{\partial p}{\partial s} \delta s\right)\alpha$, in the direction of motion. Then the resultant forward thrust is $-\frac{\partial p}{\partial s} \delta s \alpha$.

Again, let the component of the body force in the direction of motion be F , the total body force on the element of mass is $F\alpha\delta s\rho$.

Then if q be the velocity, the equation of motion is given by Newton's 2nd law,

$$\rho\alpha\delta s \frac{Dq}{Dt} = F\rho\alpha\delta s - \frac{\partial p}{\partial s} \delta s \alpha$$

$$\rho\alpha\delta s \frac{Dq}{Dt} = -\frac{\partial V}{\partial s} \rho\alpha\delta s - \frac{\partial p}{\partial s} \delta s \alpha$$

$$\text{or, } \frac{Dq}{Dt} = -\frac{\partial V}{\partial s} - \frac{1}{\rho} \frac{\partial p}{\partial s}$$

$$\text{or, } \frac{\partial q}{\partial t} + q \frac{\partial q}{\partial s} = -\frac{\partial V}{\partial s} - \frac{1}{\rho} \frac{\partial p}{\partial s}$$

$$\text{or, } q \frac{\partial q}{\partial s} = -\frac{\partial V}{\partial s} - \frac{1}{\rho} \frac{\partial p}{\partial s} \quad [\text{Since, the motion is steady } \frac{\partial q}{\partial t} = 0]$$

$$\therefore \frac{\partial}{\partial s} \left(\int \frac{dp}{\rho} + \frac{1}{2} q^2 + V \right) = 0$$

So that, constant along a stream line.

Note: If the motion is irrotational, velocity potential exists. In this case c is absolute constant. If ρ is constant then the equation (1) reduces to, $\frac{p}{\rho} + \frac{1}{2} q^2 + V = c$

Bernoulli's theorem (for barotropic flow)

Theorem: Deduce the Bernoulli's theorem for compressible fluid (Barotropic flow).

Statement: The flow will be called barotropic when the pressure is a function of density. i.e. $p = f(\rho)$.

Proof: Let us assume a steady barotropic flow and a conservation field of force for which the potential energy Ω , the energy K per unit mass is

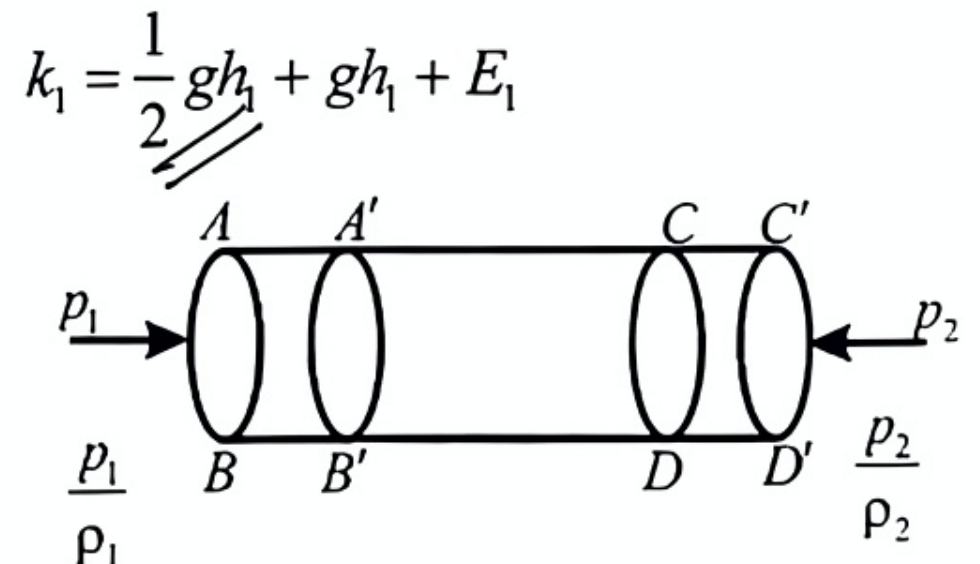
$$K = \frac{1}{2} q^2 + gh + E$$

where E is the internal energy per unit mass.

The Bernoulli's theorem becomes

$$\frac{p_1}{\rho_1} + \frac{1}{2} q_1^2 + gh_1 + E_1 = \frac{p_2}{\rho_2} + \frac{1}{2} q_2^2 + gh_2 + E_2$$

$$\text{i.e. } \frac{p_1}{\rho_1} + k_1 = \frac{p_2}{\rho_2} + k_2$$



$$k_2 = \frac{1}{2} q_2^2 + gh_2 + E_2$$

Now we find the value of internal energy.

Let S and S' be the surface of a unit mass of gas before and after a small expansion.

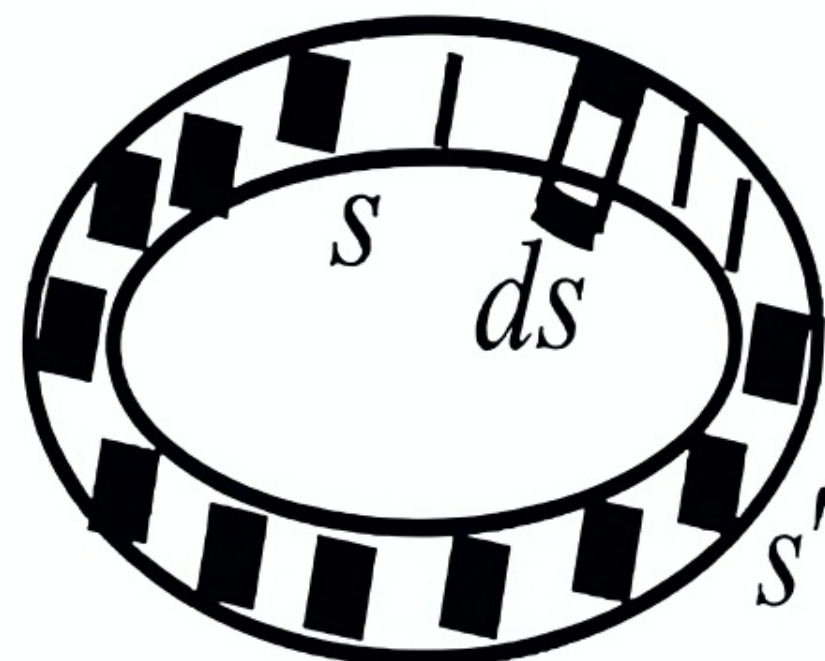
Let the normal displacement of the elements dS of the surface S be dn .

Suppose the pressure of the gas be p . Then the

work done by gas is $p \sum dn.dS = pX$ increase of

volume = $p dv$

So, the volume of the unit mass is $\rho v = 1 \therefore v = \frac{1}{\rho}$



Hence the work done by gas = $pd\left(\frac{1}{\rho}\right)$

$$\therefore e = pd\left(\frac{1}{\rho}\right)$$

$$\therefore \text{The total work done, } E = \int pd\left(\frac{1}{\rho}\right)$$

$$E = \int pd\left(\frac{1}{\rho}\right) = p \int d\left(\frac{1}{\rho}\right) - \int \left[\frac{dp}{d(1/\rho)} \int d\left(\frac{1}{\rho}\right) \right] d\left(\frac{1}{\rho}\right) = \frac{p}{\rho} - \int \frac{dp}{\rho}$$

Let the gas be expand from one step $(p \rho)$ to $(p_0 \rho_0)$

$$E = \int_{p_0}^p pd\left(\frac{1}{\rho}\right) = \frac{p_0}{\rho_0} - \frac{p}{\rho} - \int_{p_0}^p \frac{dp}{\rho} \quad \therefore E = \frac{p_0}{\rho_0} - \frac{p}{\rho} + \int_{p_0}^p \frac{dp}{\rho}$$

$$\therefore E_1 = \frac{p_0}{\rho_0} - \frac{p_1}{\rho_1} + \int_{p_0}^{p_1} \frac{dp}{\rho}$$

$$\therefore E_2 = \frac{p_0}{\rho_0} - \frac{p_2}{\rho_2} + \int_{p_0}^{p_2} \frac{dp}{\rho}$$

So, the equation becomes

$$\frac{p_0}{\rho_0} - \frac{p_1}{\rho_1} + \int_{p_0}^{p_1} \frac{dp}{\rho} + \frac{1}{2}q_1^2 + gh_1 + \frac{p_1}{\rho_1} = \frac{p_2}{\rho_2} + gh_2 + \frac{1}{2}q_2^2 + \frac{p_0}{\rho_0} - \frac{p_2}{\rho_2} + \int_{p_0}^{p_2} \frac{dp}{\rho}$$

$$\therefore \int_{p_0}^{p_1} \frac{dp}{\rho} + \frac{1}{2}q_1^2 + gh_1 = \int_{p_0}^{p_2} \frac{dp}{\rho} + \frac{1}{2}q_2^2 + gh_2$$

$$\therefore \int_{p_0}^{p_1} \frac{dp}{\rho} + \frac{1}{2}q^2 + gh = \text{Constant along a stream line in barotropic flow.}$$

Application of Bernoulli's theorem to adiabatic expansion:

Statement: The relation connects when a gas expands adiabatically the pressure and the density. $p = k\rho^r$.

From Bernoulli's theorem for barotropic flow.

$$\int_{p_0}^p \frac{dp}{\rho} + \frac{1}{2}q^2 + gh = \text{Constant.}$$

$$\begin{aligned} \text{Now } \int_{p_0}^p \frac{dp}{\rho} &= \int_{\rho_0}^{\rho} \frac{kr \rho^{r-1}}{\rho} d\rho = \int kr \rho^{r-2} d\rho \quad [i.e. p = k \rho^r \therefore dp = kr \rho^{r-1} d\rho] \\ &= kr \int \rho^{r-2} d\rho = kr \frac{\rho^{r-2+1}}{r-2+1} = \frac{kr}{r-1} \rho^{r-1} = \frac{kr}{r-1} \frac{\rho^r}{\rho} = \frac{r}{r-1} \frac{p}{\rho} = \frac{r}{r-1} \left(\frac{p}{\rho} - \frac{p_0}{\rho_0} \right) \end{aligned}$$

Since $\frac{p_0}{\rho_0}$ refers to standard state, this is constant and therefore Bernoulli's theorem gives

$$\frac{r}{r-1} \frac{p}{\rho} + \frac{1}{2} q^2 + gh = \text{Constant.}$$

If we take p_0 to be the pressure when the velocity is zero and neglects the effect of gravity

$$\therefore gh = 0$$

$$\therefore \frac{r}{r-1} \frac{p}{\rho} + \frac{1}{2} q^2 = \frac{r}{r-1} \frac{p_0}{\rho_0} \quad \therefore 2q^2 = \frac{r}{r-1} \left(\frac{p_0}{\rho_0} - \frac{p}{\rho} \right)$$

$$\therefore q^2 = \frac{2r}{r-1} \frac{p_0}{\rho_0} \left(1 - \frac{p \rho_0}{p_0 \rho} \right)$$

$$\text{Now } \frac{p \rho_0}{p_0 \rho} = \frac{k \rho^r \rho_0}{k \rho_0^r \rho} = \frac{\rho^{r-1} \rho_0}{\rho_0^{r-1}} = \left(\frac{\rho}{\rho_0} \right)^{r-1} = \left[\frac{(p/k)^{1/r}}{(p_0/k)^{1/r}} \right]^{r-1} = [p/p_0]^{\frac{r-1}{r}}$$

$$[\because p = k \rho^r, \rho^r = p/k, \rho = (p/k)^{1/r}]$$

$$\therefore q^2 = \frac{2r}{r-1} \frac{p}{p_0} \left[1 - \frac{p}{p_0} \right]^{\frac{r-1}{r}}$$

Example 4. A stream in horizontal pipe after passing a contraction in the pipe after passing a contraction in the pipe at which the sectional area A is delivered at atmospheric pressure at a place where the sectional area is B . Show that if a side tube is connected with the pipe at the formed place, Water will be sucked up through it into the pipe, S being the delivery per second.

Solution: Let ρ be the density p_1 & q_1 be the atmospheric pressure and respectively at A and p_2 & q_2 be the those of B .

By Bernoulli's theorem

$$\text{or, } \frac{p_1}{\rho} + \frac{1}{2}q_1^2 = \frac{p_2}{\rho} + \frac{1}{2}q_2^2 \quad (1)$$

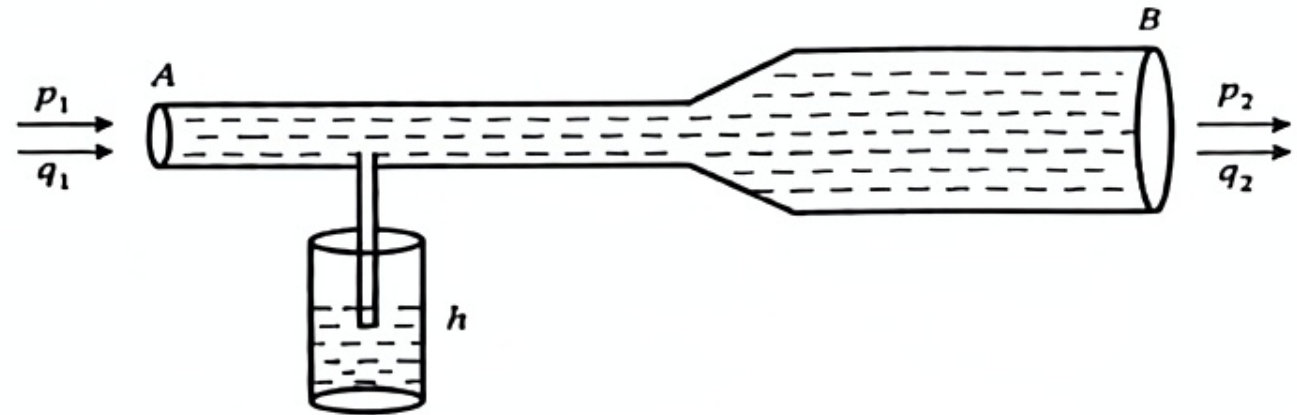
$$\text{or, } \frac{p_1 - p_2}{\rho} = \frac{1}{2}(q_2^2 - q_1^2) \quad (2)$$

Also, by equation of continuity

$$q_1 B = q_2 A = s \therefore q_1 = \frac{s}{B}, q_2 = \frac{s}{A}$$

From equation (2) we get,

$$\frac{p_1 - p_2}{\rho} = \frac{1}{2} \left(\frac{s^2}{A^2} - \frac{s^2}{B^2} \right) \quad (3)$$



Let, h be the depth of the reservoir

$$p_1 - p_2 = \rho gh$$

$$gh = \frac{p_1 - p_2}{\rho}$$

Equation (3) becomes

$$\text{or, } gh = \frac{1}{2} \left(\frac{s^2}{A^2} - \frac{s^2}{B^2} \right)$$

$$\text{or, } h = \frac{s^2}{2g} \left(\frac{1}{A^2} - \frac{1}{B^2} \right)$$

where, h is the depth of the reservoir

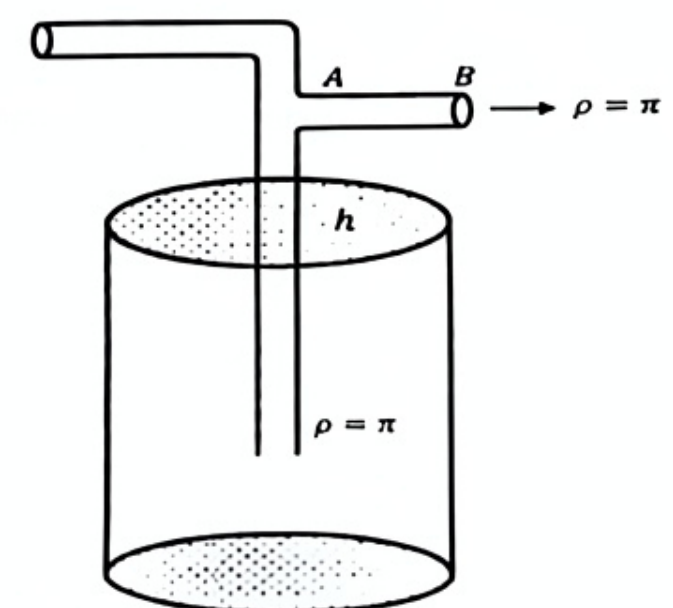
EX-1. Another Process:

Solution: The atmospheric pressure at water surface and at B is $p = \pi$

And the speed at A & B are q_1 & q_2 respectively.

Let h be the equation of continuity

$$\rho q_1 A = \rho q_2 B \quad \text{or, } q_1 A = q_2 B$$



By applying Bernoulli's theorem; $\frac{p}{\rho} + \frac{1}{2}q_1^2 + 0 = \frac{p}{\rho} + \frac{1}{2}q_2^2 + gh$

$$\therefore gh = \frac{1}{2}q_1^2 - \frac{1}{2}q_2^2 \Rightarrow h = \frac{1}{2g}(q_1^2 - q_2^2)$$

$$\therefore q_1 A = q_2 B = s$$

$$q_1 = \frac{s}{A}, q_2 = \frac{s}{B}$$

$$\therefore h = \frac{1}{2g} \left(\frac{s^2}{A^2} - \frac{s^2}{B^2} \right)$$

$$\therefore h = \frac{s^2}{2g} \left(\frac{1}{A^2} - \frac{1}{B^2} \right) \quad (\text{Proved})$$

Exercise 6. Liquid of density ρ is flowing along a horizontal pipe of variable cross-section and the pipe is connected with a differential pressure gauge at two points A & B . Show that if $p_1 - p_2$ is the pressure indicated by the gauge, the mass m of liquid flowing

through the pipe per second is given by $m = \sigma_1 \sigma_2 \sqrt{\frac{2\rho(p_1 - p_2)}{\sigma_1^2 - \sigma_2^2}}$ where σ_1 & σ_2 are the cross-sections at A & B respectively.

Solution: Let q_1 & q_2 be the speeds at A & B respectively. p_1 & p_2 are the pressure at A & B cross-section area are σ_1 & σ_2 and density ρ .

Now the conservation law, we get

$$\sigma_1 q_1 \rho = \sigma_2 q_2 \rho = m \quad (1)$$

Since $\sigma_1 q_1 t = v$

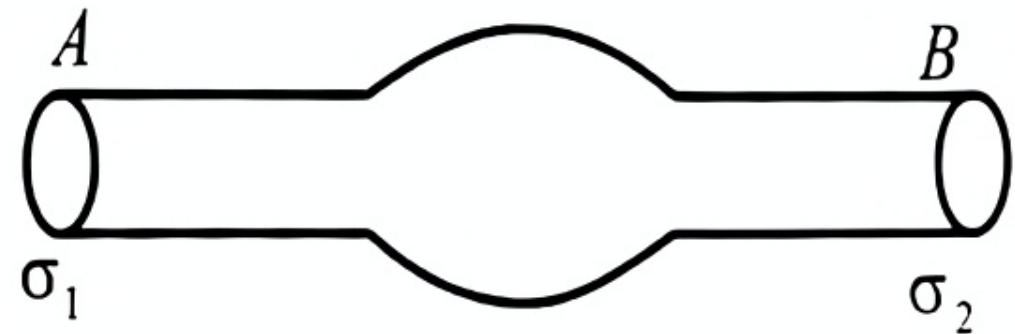
Unit time

$$\sigma_1 q_1 = v$$

$$\left[m = \sigma_1 q_1 \rho, \quad m = v \rho, \quad \rho = \frac{m}{v} \right]$$

Again, from Bernoulli's equation

$$\frac{p_1}{\rho} + \frac{1}{2}q_1^2 = \frac{p_2}{\rho} + \frac{1}{2}q_2^2 \quad \text{or,} \quad \left(\frac{p_1}{\rho} - \frac{p_2}{\rho} \right) = \frac{1}{2}(q_2^2 - q_1^2)$$



$$\text{or, } (q_2^2 - q_1^2) = 2 \left(\frac{p_1 - p_2}{\rho} \right) \quad (2)$$

From equations (1) and (2) we have

$$\begin{aligned} q_1 &= \frac{m}{\sigma_1 \rho}, \quad q_2 = \frac{m}{\sigma_2 \rho} \\ \therefore \left(\frac{m^2}{\sigma_1^2 \rho^2} - \frac{m^2}{\sigma_2^2 \rho^2} \right) &= 2 \left(\frac{p_1 - p_2}{\rho} \right) \quad \therefore \frac{m^2}{\rho^2} \left(\frac{\sigma_2^2 - \sigma_1^2}{\sigma_1^2 \sigma_2^2} \right) = 2 \left(\frac{p_1 - p_2}{\rho} \right) \\ \therefore m^2 &= \left(\frac{2 \sigma_1^2 \sigma_2^2 \rho (p_1 - p_2)}{\sigma_2^2 - \sigma_1^2} \right) \quad \therefore m = \sqrt{\frac{2 \sigma_1^2 \sigma_2^2 \rho (p_1 - p_2)}{\sigma_2^2 - \sigma_1^2}} \end{aligned}$$

CH-820 A gas which the pressure and density are connected by the adiabatic relation

$p = k \rho^\gamma$. Flows along a pipe prove that $q^2 + \frac{2\gamma}{\gamma+1} \frac{p}{\rho}$ is constant, if the pipe converges in

the direction of the flow. Prove that q will increase and $\frac{p}{\rho}$ will diminish in the direction

of the flow provided that $q^2 \rho < \lambda p$.

Solution:

We have, $p = k \rho^\gamma$

$$\therefore dp = k \gamma \rho^{\gamma-1} d\rho \quad (1)$$

But we have the Bernoulli's theorem for the compressible gases

$$\int \frac{dp}{\rho} + \frac{1}{2} q^2 + gt = \text{constant}$$

$$\text{Now, } \int \frac{dp}{\rho} + \frac{1}{2} q^2 = c$$

$$\int \frac{k \gamma \rho^{\gamma-1}}{\rho} + \frac{1}{2} q^2 = c \text{ by (1)}$$

$$\text{or, } \frac{k\gamma\rho^{\gamma-1}}{\rho(\gamma-1)} + \frac{1}{2}q^2 = c$$

$$\text{or, } \frac{\gamma}{(\gamma-1)} \frac{p}{\rho} + \frac{1}{2}q^2 = c$$

$$\text{or, } q^2 + \frac{2\gamma}{(\gamma-1)} \frac{p}{\rho} = \text{constant} \quad (2)$$

$$\text{Second part (2) we have; } q^2 = c - \frac{2\gamma}{\gamma-1}$$

$$\text{Now, } q^2 < p_0 \text{ where } c_0 = \frac{\gamma p}{\rho}$$

$$\therefore q^2 < \frac{\gamma p}{\rho} \text{ or, } q^2 \rho < \gamma p \text{ (Proved)}$$

Exercise 10. Show that the speed q of gas flowing in a thin tube whose cross-section is σ at a point of distance S in arc from a fixed cross-section obeys the equation

$$\frac{d}{dS}(\log \sigma) + \left(1 - \frac{q^2}{S^2}\right) \frac{d}{dS}(\log q) = 0$$

where c is the speed of sound in the adiabatic law being followed through out.

Solution: From the law of conservation of mass $p\sigma q = \text{constant}$ (equation of continuity).

Taking log on both sides and differentiate w.r.t S and we get

$$\log \rho + \log \sigma + \log q = \log c$$

$$\text{or, } \frac{1}{\rho} \frac{d\rho}{dS} + \frac{1}{\sigma} \frac{d\sigma}{dS} + \frac{1}{q} \frac{dq}{dS} = 0 \quad (1)$$

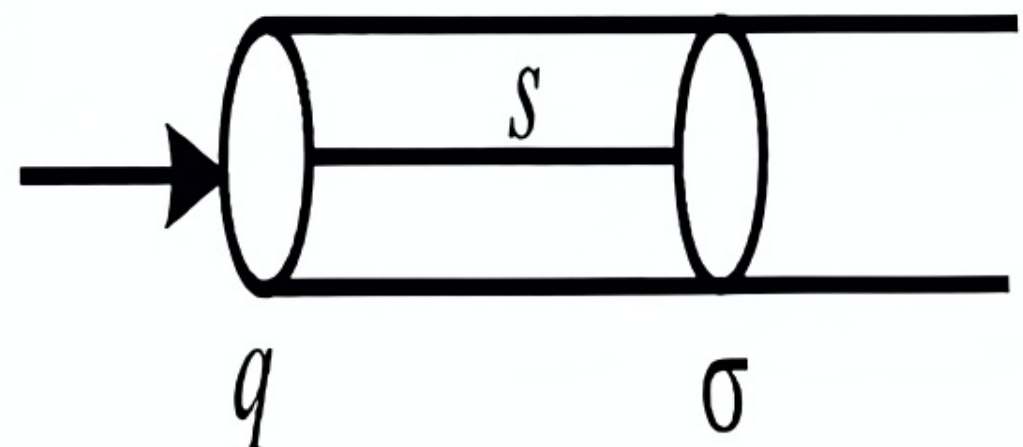
Bernoulli's equation with (adiabatic expansion)

$$\frac{\gamma}{\gamma-1} \frac{p}{\rho} + \frac{1}{2}q^2 = c$$

$$\frac{\gamma}{\gamma-1} \frac{k\rho^\gamma}{\rho} + \frac{1}{2}q^2 = c$$

$$[p = k\rho^\gamma \text{ without gain or loss of heat}]$$

Differentiating w. r. to S



$$\frac{k\gamma}{\gamma-1}(\gamma-1)\rho^{\gamma-2}\frac{d\rho}{dS} + \frac{1}{2}2q\frac{dq}{dS} = 0$$

$$\text{or, } k\gamma\rho^{\gamma-2}\frac{d\rho}{dS} + q\frac{dq}{dS} = 0 \quad \text{or, } \frac{k\gamma\rho^{\gamma}}{\rho^2}\frac{d\rho}{dS} + q\frac{dq}{dS} = 0$$

$$\text{or, } \frac{\gamma p}{\rho^2}\frac{d\rho}{dS} + q\frac{dq}{dS} = 0 \quad [\because p = k\rho^{\gamma}]$$

$$\text{or, } \frac{\gamma p}{\rho^2}\frac{1}{\rho}\frac{d\rho}{dS} + q\frac{dq}{dS} = 0$$

$$\text{or, } \frac{c^2}{\rho}\frac{d\rho}{dS} + q\frac{dq}{dS} = 0 \quad \left[\because c^2 = \frac{\gamma p}{\rho} \right]$$

$$\text{or, } \frac{1}{\rho}\frac{d\rho}{dS} + \frac{q}{c^2}\frac{dq}{dS} = 0$$

$$\text{or, } \frac{1}{\rho}\frac{d\rho}{dS} = -\frac{q}{c^2}\frac{dq}{dS} \quad (2)$$

From equations (1) and (2) we get

$$-\frac{q}{c^2}\frac{dq}{dS} + \frac{1}{\sigma}\frac{d\sigma}{dS} + \frac{1}{q}\frac{dq}{dS} = 0$$

$$\text{or, } \left(\frac{1}{q} - \frac{q}{c^2} \right) \frac{dq}{dS} + \frac{1}{\sigma} \frac{d\sigma}{dS} = 0$$

$$\text{or, } \frac{1}{q} \left(1 - \frac{q^2}{c^2} \right) \frac{dq}{dS} + \frac{1}{\sigma} \frac{d\sigma}{dS} = 0$$

$$\text{or, } \frac{1}{q} \left(1 - \frac{q^2}{c^2} \right) \frac{dq}{dS} + \frac{1}{\sigma} \frac{d\sigma}{dS} = 0$$

$$\therefore \left(1 - \frac{q^2}{c^2} \right) \frac{d}{dS} (\log q) + \frac{d}{dS} (\log \sigma) = 0$$

Exercise 11. If gas flows from a vessel through a small orifice from a region where the pressure is p_1 to a region and orifice is p_2 . Prove that the rate of efflux of mass is

$$c_2 \rho_2 w_2 \sqrt{\frac{2}{\gamma-1} \left[\left(\frac{p_2}{p_1} \right)^{1/\gamma-1} - 1 \right]} \quad \text{where } p = k \rho^\gamma, w_2 \text{ is the area of the vena contracta and } c_2$$

is the speed of sound in the gas, $c_2^2 = \frac{\gamma p^2}{\rho_2}$, ρ_2 being the density of the vena contracta.

Solution: The equation of conservation of mass is $w_1 q_1 \rho_1 = w_2 \rho_2 q_2$ where each term indicated the rate of flow of mass. We have by Bernoulli's equation for adiabatic flow

$$\frac{\gamma p}{(\gamma-1)\rho} + \frac{1}{2} q^2 = c \text{ at the cross-section } w_1, p = p_1, \rho = \rho_1, q = q_1$$

$$\therefore \frac{\gamma p_1}{(\gamma-1)\rho_1} + \frac{1}{2} q_1^2 = c \quad (1)$$

at the vena contracta.

$$w_2, p_2, \rho_2, q_2$$

$$\frac{\gamma p_2}{(\gamma-1)\rho_2} + \frac{1}{2} q_2^2 = c \quad (2)$$

Subtracting (1) from (2) we get

$$\begin{aligned} \frac{1}{2} (q_2^2 - q_1^2) &= \frac{\gamma}{(\gamma-1)} \left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1} \right) \\ &= \frac{\gamma}{(\gamma-1)} \frac{p_2}{\rho_2} \left(1 - \frac{\rho_2 p_1}{p_2 \rho_1} \right) = \frac{c_2^2}{(\gamma-1)} \left(1 - \frac{\rho_2 p_1}{p_2 \rho_1} \right) \end{aligned}$$

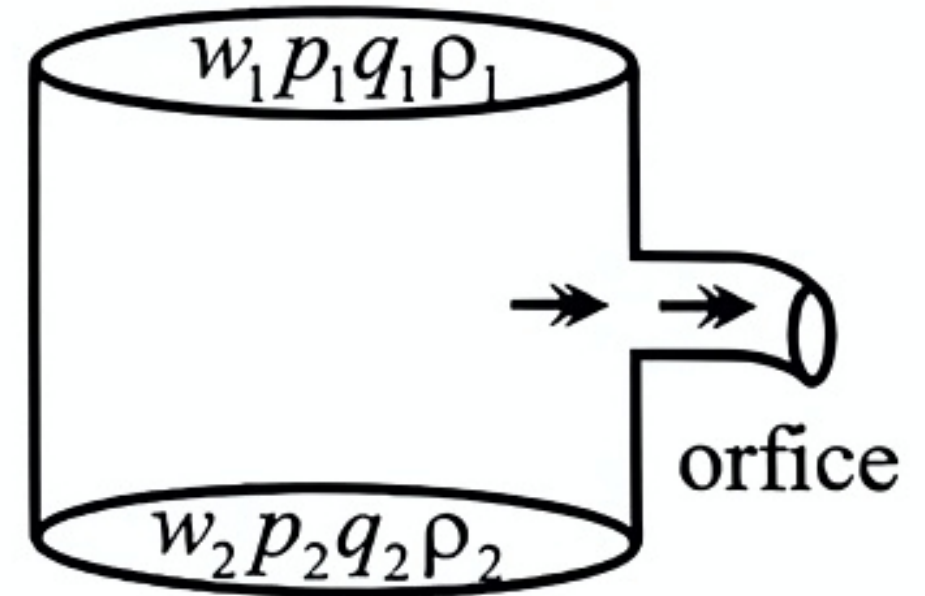
$$\left[c_2^2 = \frac{\gamma p_2}{\rho_2}, p_1 = k \rho_1^\gamma, p_2 = k \rho_2^\gamma, \frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma \right]$$

$$= \left(\frac{1}{p_2 / p_1} \times \frac{\rho_2}{\rho_1} - 1 \right) = \left\{ \frac{1}{p_2 / p_1} \times \left(\frac{p_2}{p_1} \right)^{1/\gamma} - 1 \right\}$$

$$= \left\{ \left(\frac{p_2}{p_1} \right)^{1/\gamma-1} - 1 \right\}$$

$$\therefore q_2^2 - q_1^2 = \frac{2c_2^2}{\gamma-1} \left\{ \left(\frac{p_2}{p_1} \right)^{1/\gamma-1} - 1 \right\}$$

$$\text{Since } q_2 \text{ is very large in compare with } q_1 \text{ i.e. } q_1 \rightarrow 0 \therefore q_2 = \sqrt{\frac{2c_2^2}{\gamma-1} \left\{ \left(\frac{p_2}{p_1} \right)^{1/\gamma-1} - 1 \right\}}$$



Exercise 14. Gases flow radially from a point symmetrically in all direction pressure and density being connected by the law $p = k\rho$. If m is the rate of emission of mass supposed constant prove that $4\pi qr^2 = \max p \left[\frac{q^2 - q_1^2}{2k} \right]$ where q is the speed Now after

the time $(t + \delta t)$ the fluid will reach the surface of a sphere of radius r is $\frac{4\pi r^2 q \delta t \rho}{\delta t} = m$

or, $m = 4\pi r^2 q \rho$ [$m = \text{mass}$]

Now general equation of Bernoulli's theorem is

$$\int \frac{dp}{\rho} + \frac{1}{2} q^2 = c \text{ (Neglecting external force due to gravity)}$$

We have, $p = k\rho$

$$dp = k d\rho$$

$$\int \frac{dp}{\rho} + \frac{1}{2} q^2 = c \text{ becomes}$$

$$\int \frac{k d\rho}{\rho} + \frac{1}{2} q^2 = c$$

$$k \log(\rho) + \frac{1}{2} q^2 = c \quad (1)$$

when $p = 1$, $q = 1$ then the equations become,

$$k \log(1) + \frac{1}{2} q^2 = c \quad (\because \log 1 = 0)$$

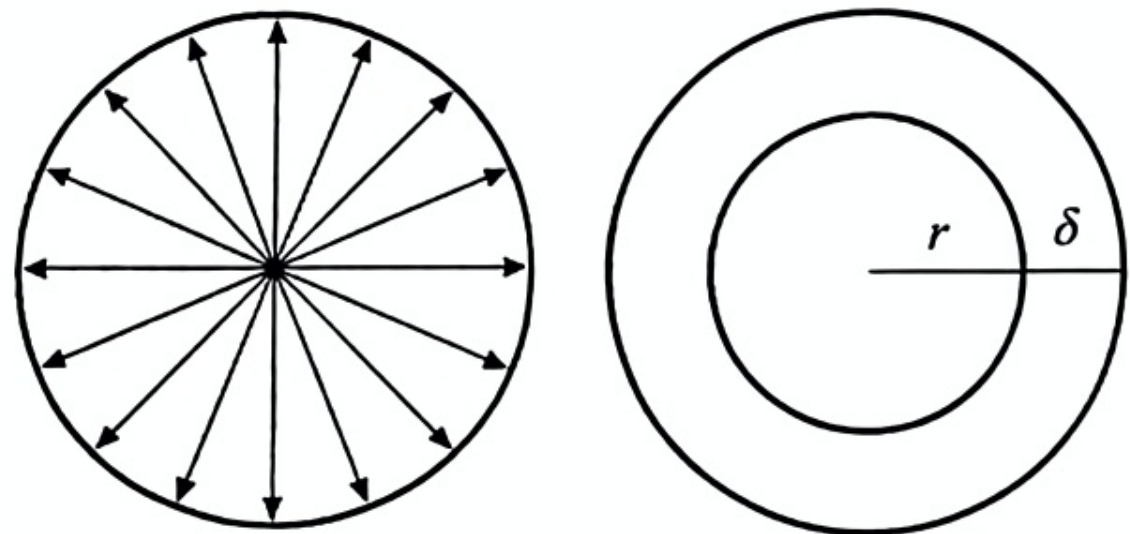
$$\text{or, } k \log(\rho) + \frac{1}{2} q^2 = \frac{1}{2} q_1^2$$

$$\text{or, } \log(\rho) = \frac{q_1^2 - q^2}{2k}$$

$$\text{or, } \rho = e^{\left(\frac{q_1^2 - q^2}{2k} \right)}$$

$$\text{Now, } m = 4\pi r^2 q \rho$$

$$\text{or, } 4\pi r^2 q e^{\left(\frac{q_1^2 - q^2}{2k} \right)} = m$$



$$\text{or, } 4\pi r^2 q = m e^{\left(\frac{q_1^2 - q^2}{2k}\right)}$$

$$\text{or, } 4\pi r^2 q = m \exp\left(\frac{q_1^2 - q^2}{2k}\right) \text{ (Proved)}$$

Exercise 8. If $\frac{p}{\rho^\gamma} = \text{constant}$ and the fluid flows out through in which the pressure is n times the atmospheric pressure p . Show that the speed v of efflux is given by

$$v^2 = \frac{2\gamma p}{(\gamma - 1)\rho} \left[n^{\left(1 - \frac{1}{\gamma}\right)} - 1 \right]$$

Solution:

$$\frac{p}{\rho^\gamma} = \text{constant}$$

$$\text{or, } p = k \rho^\gamma$$

The pressure and the density relation show that the flow adiabatic.

Here Bernoulli's theorem for the adiabatic flow applies

$$\frac{\gamma}{\gamma - 1} \frac{p}{\rho} + \frac{1}{2} q^2 = \text{constant} \quad (1)$$

Within the vessel the speed of flow is very small compared with the speed at the vena contracts

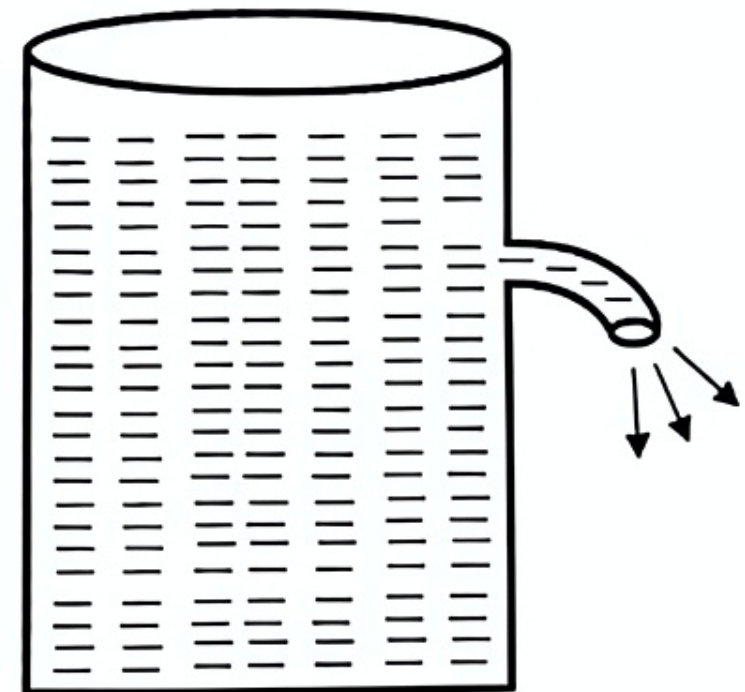
$$\frac{\gamma}{\gamma - 1} \frac{np}{\rho} + \frac{1}{2} 0^2 = c \Rightarrow \frac{\gamma}{\gamma - 1} \frac{np}{\rho} = c$$

If v be the speed at the vena contracts respectively

$$\text{we have, } \frac{\gamma}{\gamma - 1} \frac{np}{\rho} + \frac{1}{2} v^2 = \frac{\gamma}{\gamma - 1} \frac{np}{\rho}$$

$$\text{or, } \frac{1}{2} v^2 = \frac{\gamma p}{\gamma - 1} \left(\frac{n}{\rho_1} - \frac{1}{\rho} \right)$$

within the vessel and at the vena contracts respectively



$$\therefore \frac{1}{v^2} = \frac{\gamma p}{\gamma - 1} \left(\frac{np}{\rho_1} - 1 \right)$$

$$p = k \rho^\gamma \quad (2)$$

$$\text{and } np = k \rho_1^\gamma \quad (3)$$

Dividing (2) & (3)

$$\frac{1}{n} = \frac{\rho^\gamma}{\rho_1^\gamma} \Rightarrow \frac{\rho}{\rho_1} = \left(\frac{1}{n} \right)^{\frac{1}{\gamma}}$$

$$v^2 = \frac{2\gamma p}{(\gamma - 1)\rho} \left[n^{\left(1 - \frac{1}{\gamma}\right)} - 1 \right] \quad (\text{Proved})$$

For Assignment

Art 1.43 Subsonic and supersonic flow

Art 1.64 Flow of gas in converging pipe

Art 1.70 The venture tube

Art 1.93 The flow past an obstacle

Exercise No. 8 and 9